

Status Note: Symplectic Codimension-One Compact Perturbations

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Status. `literature_already_answered`. This is a compact literature-status note, not a new proof packet.

Source Question

The source paper is J. M. F. Castillo, W. Cuellar, M. Gonzalez Ortiz, and R. Pino, *Symplectic forms on Banach spaces*, arXiv:2204.01703. In the published numbering quoted by the later paper, the relevant source result is Proposition 8.11. In the arXiv PDF it appears on page 20.

It states that if $\mathcal{H}$ is a real infinite-dimensional Hilbert space and ω is a symplectic structure on $\mathcal{H}$, then there do not exist bounded bilinear maps Ω and κ on $X = \mathbb{R} \oplus \mathcal{H}$ such that κ is compact, Ω extends ω , and $\Omega + \kappa$ is symplectic on X . Immediately after the proof, the source asks whether this proposition holds for general Banach spaces.

Supporting Answer

The supporting paper is H. Li and C. Zhu, *Skew-adjoint linear relations between Banach spaces*, arXiv:2404.08445. Its Introduction, page 4, explicitly quotes the source question after Proposition 8.11 and says that the paper will solve the problem by generalizing mod-2 kernel stability to real Banach spaces. The relevant general stability statements are Theorems 1.3 and 1.4.

The direct answer is Corollary 1.5, page 5. It states that if (X, ω) and $(X \oplus \mathbb{R}^n, \tilde{\omega})$ are real symplectic Banach spaces and $L_{\tilde{\omega}|X} - L_{\omega}$ is compact, then, in the strong symplectic case, n must be even. Setting $n = 1$ rules out the codimension-one compact perturbation scenario in the source question.

Conclusion and Scope

The source question is already answered in the later literature. Proposition 8.11 does extend to general Banach spaces in the strong symplectic setting: a strong real symplectic Banach space cannot become a codimension-one strong real symplectic Banach extension by a compact perturbation of the restricted symplectic operator. The supporting paper proves a stronger parity theorem for skew-adjoint Fredholm relations; this packet records the explicit source-to-supporting-paper identification.

Local Evidence

The packet contains the original arXiv PDF as `source_paper.pdf` and the supporting answer as `supporting_paper_2404.08445.pdf`. Cheap run-index search for the arXiv id, title keywords,

symplectic Banach spaces, Proposition 8.11, and compact perturbation found no prior solution packet for this source target.

References

- [1] J. M. F. Castillo, W. Cuellar, M. Gonzalez Ortiz, and R. Pino, *Symplectic forms on Banach spaces*, arXiv:2204.01703.
- [2] H. Li and C. Zhu, *Skew-adjoint linear relations between Banach spaces*, arXiv:2404.08445.